

ASSIGNMENT 2

AHCI

NOTE:

Max marks= 50

Clearly stick to HCI concepts, considering the book and related materials

Late submission would result in zero marks

Assignment should be submitted online via GCR only

Q1) What is a **hot spot** in the context of pointer-based interaction? Why is it important in user interface design? Illustrate your answer with a suitable real-world example.

Your expected answer must be centered on the concepts of accuracy, predictability, usability, and consistency.

Q2) Ali is a university student working on a report using a word processing application. Initially, he uses the mouse to perform actions like saving the document, copying text, and pasting content. Over time, he learns keyboard shortcuts such as Ctrl + S, Ctrl + C, and Ctrl + V.

After using these shortcuts, Ali notices that he can complete his work faster and with less effort.

Question:

Based on the above scenario, explain the concept of keyboard accelerators. How do they improve user efficiency and interaction? Also, mention one possible limitation of using keyboard accelerators.

Q3) In HCI, 'augmenting man's intellect' has been an important concept for designing complex computer systems like word processors and mouse but it has been noted that these inventions gained significant attention after a long time. Describe the cause of the delay that occurred, considering Engelbart's conclusion.

Engelbart, founder of the Augmentation Research Centre, wrote of how humans attack complex intellectual problems like carpentry and woodwork. He concluded that the secret behind producing a physical computing module originating from human intellect lies in an entity. Describe that entity.

Q4) Discuss how the full- page word processor is or is not a direct manipulation interface for editing a document using Schneiderman's criteria. What features of a modern word processor break the metaphor of composition with pen (or typewriter), and paper?

Q5) Why are cultural probes considered useful for private and sensitive research contexts? Provide an example. How does it differ from ethnographic research?

Design a cultural probe package for the following topic:

“Understanding how 20-30 year olds manage their physical health habits”.